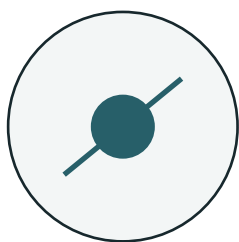


Neuro motility and emergency localisation

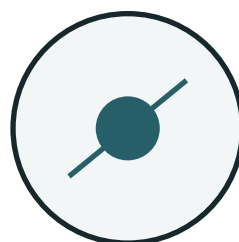
Part 2 | Skew: vertical alignment plus central context

A vertical deviation is not automatically an isolated fourth-nerve palsy.

Hypertropic: intorsion



Hypotropic: extorsion



Ocular tilt reaction: vertical misalignment plus conjugate torsion, often with central signs.

Original ocular tilt sketch: hypertropic intorsion and hypotropic extorsion.

Examine torsion, head posture, gaze behaviour and associated neurological signs. Skew can accompany imbalance in vestibular pathways, while a trochlear lesion has a different mechanical pattern. No single bedside manoeuvre perfectly excludes a central cause. Explain how the observed pattern changes urgency and the need for neurological assessment rather than relying on one alignment measurement.

Trochlear pathway: crossing changes laterality

Nuclear and peripheral lesions predict different sides.

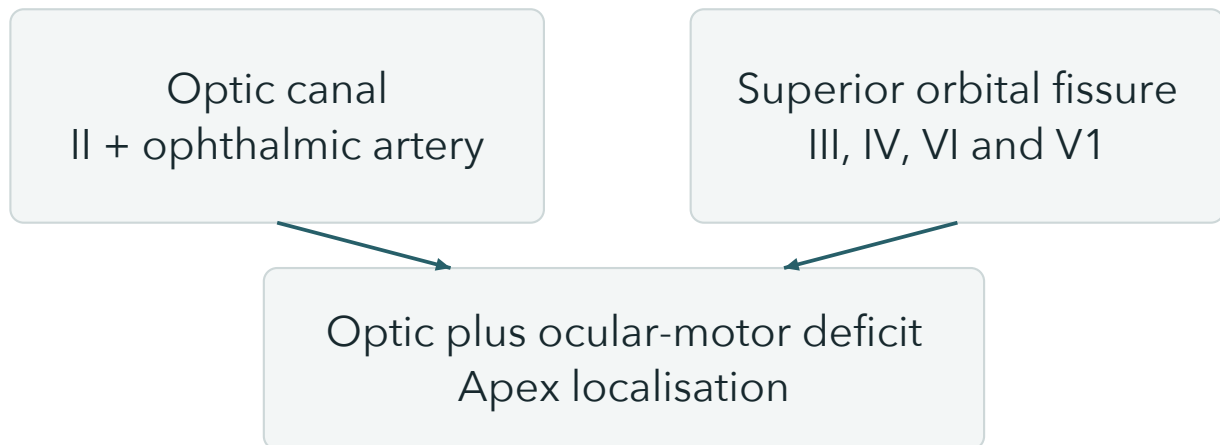
Finding	Implication
Trochlear nucleus	Axons cross before dorsal exit
Unilateral nuclear lesion	Contralateral superior-oblique weakness
Peripheral orbital nerve lesion	Ipsilateral superior-oblique weakness
Bilateral findings	Consider extent and associated midbrain signs

Name where the fibres cross before assigning laterality. A lesion of the left nucleus differs from a lesion of the left orbital nerve; the same word fourth nerve is insufficient. Link the expected weak muscle to depression in adduction, torsion and compensatory posture, then ask whether adjacent structures explain other examination findings.

Orbital apex: afferent and motor functions together

Optic dysfunction distinguishes involvement beyond the fissure.

Adjacency creates a syndrome. Examine optic function as well as eye movements.



Original anatomical adjacency: optic canal and superior orbital fissure.

Document acuity, colour, fields and the afferent pupil response alongside eye movements and corneal sensation. Combined optic dysfunction and ophthalmoplegia supports an apex pattern; preserved vision does not settle every lesion at an early stage. Imaging must address the suspected compartment and the clinical tempo, with infection, inflammation and mass lesions considered in context.

Pituitary apoplexy: stabilise before elegant localisation

Haemodynamic compromise makes cortisol replacement urgent.

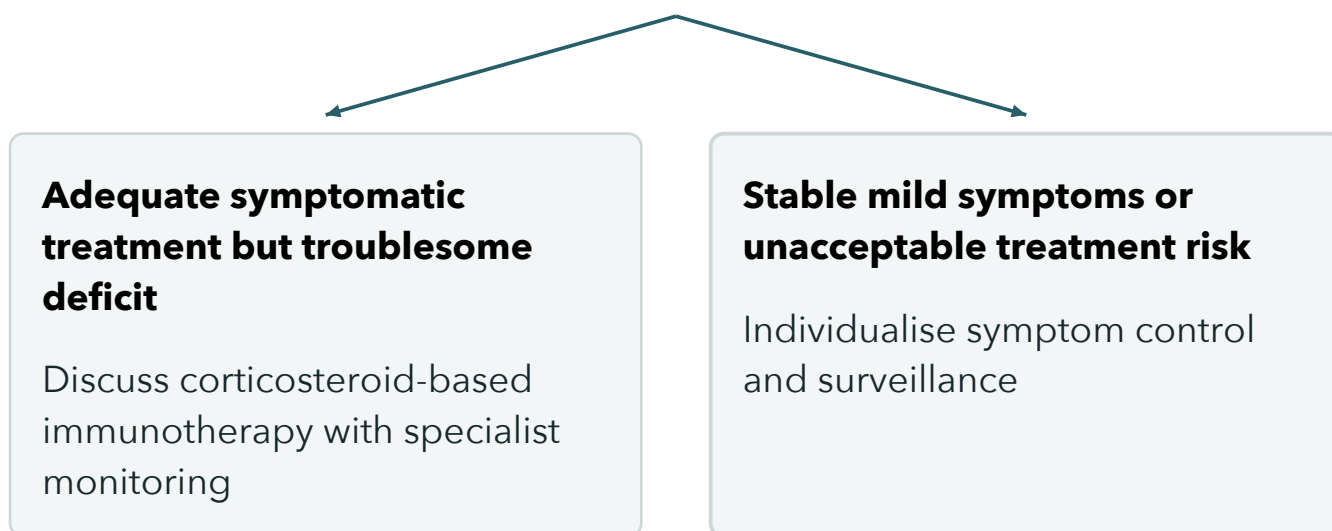
- Recognise acute headache with visual or ocular-motor change
↓
- Assess haemodynamics, consciousness and electrolytes
↓
- Give indicated empirical hydrocortisone without unsafe delay
↓
- Coordinate endocrine and neurosurgical evaluation

The systemic emergency is not secondary to the field defect. Hypotension, reduced consciousness or major visual compromise can justify immediate empirical steroid treatment while investigations proceed. Obtain relevant bloods where feasible without delaying stabilisation. Discuss decompression and longer-term endocrine care through the specialist team rather than treating the case as an isolated pupil or motility exercise.

Ocular myasthenia: disabling symptoms despite initial therapy

Immunotherapy selection differs from a positive-test exercise.

Is ocular weakness still functionally limiting?



Screen immediately for bulbar or respiratory involvement whenever the phenotype changes. Consensus guidance supports immunosuppression for functionally limiting ocular disease inadequately controlled by anticholinesterase treatment. Thymectomy is not a universal ocular first step; selected AChR-positive patients may merit discussion in particular circumstances. Explain the treatment objective, adverse effects and reassessment plan.